1. PROBLEM & GOAL

Solari Analytics handles ~150 support tickets daily, and agents frequently waste time re-solving old issues. Goal: Build a lightning-fast retrieval system that instantly surfaces top-matching past resolutions right when an agent opens a new ticket.

2. KEY ASSUMPTIONS

- Similarity definition: Blends technical keywords (e.g., "Okta", "500 error"), categories, and the core problem description.
- Strict SLA (<50ms): Must load instantly with zero UI lag.
- Scalability: Designed to grow from ~4,500 monthly tickets to over 50,000 annual records.
- Graceful fallback: If an issue is totally brand new, the system explicitly returns "No Relevant Match Found" rather than forcing low-quality results.

3. TECHNICAL APPROACH (V1)

We are using a deterministic TF-IDF + Cosine Similarity engine:

- Document Building: Combines the ticket subject, description, category, and tags into a single normalized text string.
- Vectorization: Extracts unigrams and bigrams using TF-IDF weighting while dropping generic stop words.
- Scoring & Threshold: Computes cosine similarity between the incoming query and history, using a 0.20 minimum cutoff to filter out weak matches.

4. TRADE-OFF ANALYSIS

- Naive Keyword Overlap: Blazing fast (<1ms), but too inaccurate because it treats all words equally.
- TF-IDF + Cosine Similarity (Chosen): Super fast (<5ms), highly accurate for tech jargon, zero cost, and completely free of hallucination risks.
- Generative LLM / RAG: Too slow (1s+), expensive per query, and prone to hallucinating incorrect troubleshooting steps.

5. FUTURE ROADMAP (V2+)

- Hybrid Search: Layer in lightweight local embeddings (e.g., all-MiniLM-L6-v2) to catch conceptual synonyms alongside exact keywords.

- Database Scaling: Move from in-memory processing to PostgreSQL (pgvector) or Elasticsearch once we cross 50k historical records.

- Feedback Loop: Let agents give a simple thumbs up/down to continuously tune match accuracy over time.